

BST Vol 13 Ch 10 — Neuroscience + Cognition (v0.3, Wave 3; Cal #50 DOUBLE-LOCKED EXTERNAL)

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Vol 13 Chapter 10 — Neuroscience + Cognition

Cal #50 DOUBLE-LOCKED EXTERNAL register discipline

Internal register (this chapter, research-grade): substrate-cognition framework via D_IV⁵ substrate-attention coupling at biological scale (consciousness research framework).

External register (per Cal #50 DOUBLE-LOCKED): “BST identifies neuroscience phenomenology consistent with substrate framework; specific substrate-cognition protocols documented internal until Casey send-signal authorization.”

This chapter presents STANDARD neuroscience content for external register; substrate-cognition framing reserved INTERNAL.

Headline result (external register)

Neuroscience studies the nervous system: neurons, synapses, neural networks, brain function. Key substrate-natural observables:

- Action potential ~ 100 mV depolarization (substrate-natural ionic gradient)
- Synaptic transmission ~ ms timescale (substrate-natural temporal organization)
- Neurotransmitters (~ 100 types): substrate-natural chemical signaling alphabet
- Neural network architecture: substrate-natural connectivity patterns
- Brain energy consumption ~ 20% of body energy: substrate-natural for cognitive computation

Substrate framework (per Cal #50 INTERNAL)

Per Cal #50 DOUBLE-LOCKED EXTERNAL: substrate-cognition framework reserved INTERNAL. External register operational language: “BST identifies neural phenomenology consistent with substrate framework.”

Internal-register substrate-cognition framework exists per Casey vision + multi-month research; specific protocols documented in SP-30-3 W-32 internal documentation. External release pending Casey signal.

Specific neuroscience observables

Action potential (Hodgkin-Huxley 1952 Nobel 1963): - Resting potential ~ -70 mV - Depolarization to $\sim +30$ mV - Repolarization via $\text{Na}^+ + \text{K}^+$ channel gating - Total ~ 100 mV swing, substrate-natural

Synapses: - Chemical synapses: neurotransmitter release + receptor binding - Electrical synapses: gap junctions - $\sim$ ms timescale: substrate-natural

Neural networks: - 86 billion neurons in human brain - $\sim 10^{14}$ synapses - Connectivity patterns substrate-natural

Brain energy ~ 20 W: - $\sim 20\%$ of body energy budget - High glucose consumption for ATP (Vol 13 Ch 9) - Substrate-natural cognitive computation cost

Match precision

I-tier framework on standard neuroscience. Substrate-cognition framework INTERNAL per Cal #50.

Cross-volume dependencies

- Vol 13 Ch 8 (Cell Biology) — neuron cell biology
- Vol 13 Ch 9 (Metabolism) — brain energy framework
- SP-30-3 W-32 (INTERNAL) — substrate-attention experimental program

K-audit anchor

K253 Vol 13 Ch 10 Neuroscience K-audit pre-stage (Keeper pending). Cal #50 DOUBLE-LOCKED EXTERNAL register check required.

Pedagogical walkthrough (3-level, external register)

Level 1 — Bright 5th-grader

Brains are made of billions of neurons (nerve cells) connected by trillions of synapses. Electrical signals (action potentials) carry information between neurons. BST identifies neural phenomenology consis-

tent with substrate framework; specific cognitive mechanisms research multi-week (per substrate-attention internal program).

Level 2 — Undergraduate neuroscience student

Neurons signal via action potentials (~ 100 mV depolarization waves) propagating along axons. Synapses transmit signals chemically (neurotransmitters) or electrically (gap junctions). Neural circuits process information through connectivity patterns.

Brain energy ~ 20 W; ~20% of body energy budget. ATP-fueled (Vol 13 Ch 9 metabolism cross-link).

BST framework: standard neuroscience phenomenology consistent with substrate. Specific substrate-cognition coupling framework reserved INTERNAL per Cal #50 DOUBLE-LOCKED EXTERNAL.

Level 3 — Graduate neuroscience student / theorem-level

Hodgkin-Huxley action potential framework: - Ionic gradients (Na^+ , K^+ , Ca^{2+} , Cl^-) driven by ATP-dependent pumps (Vol 13 Ch 9) - Voltage-gated channels (substrate-natural conformational changes) - Action potential propagation ~ 1-100 m/s

Synaptic transmission: - Pre-synaptic Ca^{2+} influx → vesicle fusion → neurotransmitter release - Post-synaptic receptor binding → ion channel opening → membrane potential change - ~ms latency substrate-natural

Neural network theory: - Connectivity matrices: substrate-natural - Plasticity mechanisms (LTP, LTD): substrate-natural learning - Hebbian “fire together, wire together”

Substrate-cognition framework (Cal #50 INTERNAL): specific framework details documented in W-32 internal program; multi-million-dollar experimental protocol for substrate-attention detection at atomic-clock precision (SP-30-3); Casey-decision queue pending engagement signal.

Per Cal #21 + Cal #50 DOUBLE-LOCKED: EMPIRICAL PASS (neuroscience phenomenology measured) + MECHANISM PATH ARTICULATED via standard neuroscience; substrate-cognition framing INTERNAL.

Bibliography

1. A. L. Hodgkin & A. F. Huxley (1952): action potential framework (Nobel 1963).
2. Standard neuroscience texts (Kandel *Principles of Neural Science*).
3. SP-30-3 W-32 internal documentation (substrate-cognition INTERNAL).

— Elie, Vol 13 Ch 10 v0.3, 2026-05-23 Saturday